



Generative AI Applications in SE Lab 05

Submitted to

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BSSE-VI-B

Task 1: Dynamic Similarity Threshold Implementation

1.1 Objective

Modify the Build Context code node to dynamically accept a minScore parameter in the request body, allowing users to control the retrieval confidence threshold. Include all three match scores in the response via a scores array and scoreLabels array for full retrieval transparency.

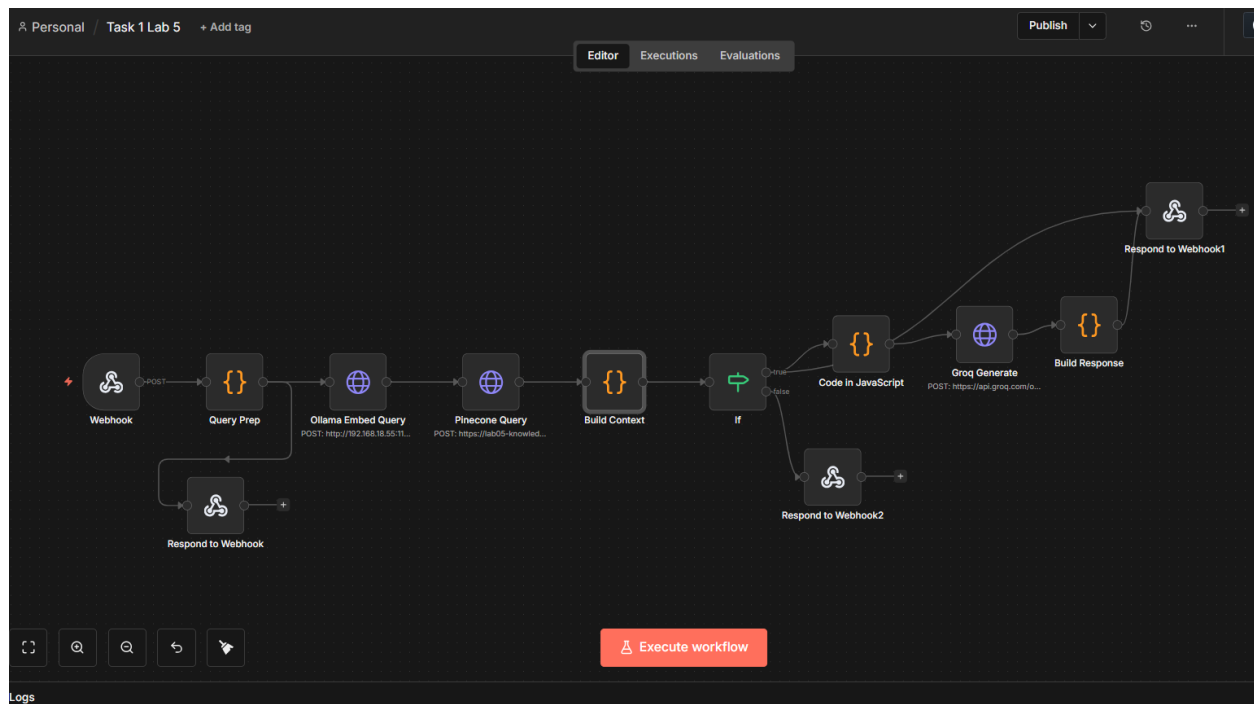
1.2 Implementation Summary

The following changes were made to the Lab05-RAG-Groq-Chatbot workflow:

- Query Prep node: Reads minScore from request body (default: 0.60) and passes it downstream.
- Build Context node: Uses dynamic minScore threshold instead of hardcoded 0.60 to filter Pinecone matches.
- Build Context node: Returns a scores array (floats) and a scoreLabels array (formatted as 'KB-XXX: 0.XXX') alongside retrieved context.
- Response: scoreLabels and scores fields added to the outgoing JSON so callers can inspect retrieval confidence.

1.3 n8n Workflow — Node Configuration

The screenshot below shows the Task 1 workflow canvas in n8n.



Build Context Node — Updated Code

The screenshot below shows the updated JavaScript inside the Build Context node with dynamic minScore logic.

```

Language
JavaScript

JavaScript

1  const pineconeResult = $input.item.json;
2  const matches = pineconeResult.matches || [];
3
4  const THRESHOLD = $input.item.json.minScore ?? 0.60;
5  const relevant = matches.filter(m => m.score >= THRESHOLD);
6
7  let contextBlock = '';

```

1.4 Test Cases & Results

Test Case 1 — High Threshold (minScore: 0.80)

Request Body	{ "question": "What is byte pair encoding?", "minScore": 0.80, "studentName": "Ehtisham" }
Expected	Only matches with score ≥ 0.80 included. retrievedCount: 2. scoreLabels shows high-confidence entries only.
Actual Status	Pass
retrievedCount	1
topScore	0.696

My Collection > Post data

POST http://localhost:5678/webhook-test/6e844034-45c3-4ea3-8221-19e965155522

Docs Params Authorization Headers 9 Body Scripts Tests Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

Schema Beautify

```

1  {
2    "question": "What is byte pair encoding?", "minScore": 0.80, "studentName": "Ehtisham"
3  }

```

Body Cookies Headers 8 Test Results 1/1

200 OK • 1.24 s • 1.11 KB • Save Response

```

1  {
2    "contextFound": true,
3    "contextBlock": "[1] Tokenisation splits raw text into units called tokens before processing by an LLM.
4    Modern models use subword tokenisation algorithms such as Byte Pair Encoding (BPE), which merge frequent
5    character pairs iteratively to build a vocabulary. This allows the model to handle rare and
6    out-of-vocabulary words by decomposing them into known subword units. (Source: Sennrich et al. (2016).
7    BPE.)\n\n",
8    "sources": [
9      "KB-015 - NLP (Sennrich et al. (2016). BPE.)"
10   ],
11   "scores": [
12     0.696
13   ],
14   "scoreLabels": [
15     "KB-015: 0.696"
16   ],
17   "topScore": 0.696,
18   "retrievedCount": 1,
19   "question": "",
20   "studentName": "",
21   "history": [],
22   "minScore": 0.6

```

Test Case 2 — Low Threshold (minScore: 0.50)

Request Body	{ "question": "What is byte pair encoding?", "minScore": 0.50, "studentName": "Ehtisham" }
Expected	More matches included (up to 3). retrievedCount: 3. All scores ≥ 0.50 retrieved.
Actual Status	Pass
retrievedCount	1____
topScore	0.696____

Overview GET Get data POST Post data No environment

My Collection > Post data Save Share

POST http://localhost:5678/webhook-test/6e844034-45c3-4ea3-8221-19e965155522 Send

Docs Params Authorization Headers 9 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

1 { "question": "What is byte pair encoding?", "minScore": 0.50, "studentName": "Ehtisham" }

Body Cookies Headers 8 Test Results 1/1 200 OK - 1.29 s - 1.11 KB Save Response

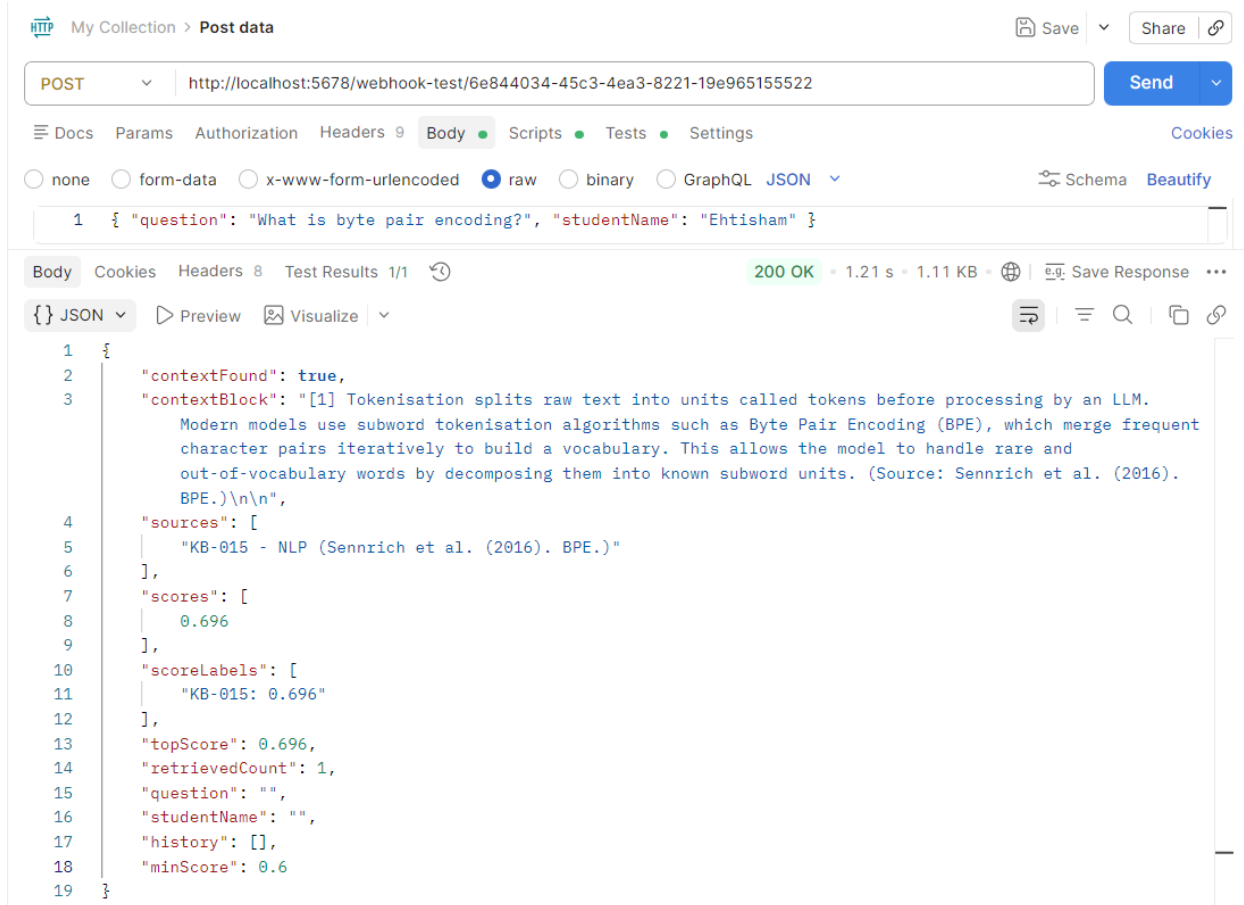
{ } JSON Preview Visualize

```

1 {
2   "contextFound": true,
3   "contextBlock": "[1] Tokenisation splits raw text into units called tokens before processing by an LLM.
    Modern models use subword tokenisation algorithms such as Byte Pair Encoding (BPE), which merge frequent
    character pairs iteratively to build a vocabulary. This allows the model to handle rare and
    out-of-vocabulary words by decomposing them into known subword units. (Source: Sennrich et al. (2016).
    BPE.)\n\n",
4   "sources": [
5     "KB-015 - NLP (Sennrich et al. (2016). BPE.)"
6   ],
7   "scores": [
8     0.696
9   ],
10  "scoreLabels": [
11    "KB-015: 0.696"
12  ],
13  "topScore": 0.696,
14  "retrievedCount": 1,
15  "question": "",
16  "studentName": "",
17  "history": [],
18  "minScore": 0.6
19 }
```

Test Case 3 — Default Threshold (No minScore field)

Request Body	{ "question": "What is byte pair encoding?", "studentName": "Ehtisham" }
Expected	Query Prep defaults to 0.60. Backward compatibility maintained. Response identical to pre-task baseline.
Actual Status	Pass
retrievedCount	1____
topScore	0.696____



Task 2: Domain Knowledge Base Expansion

2.1 Objective

Expand the Lab04-KnowledgeBase Google Sheet with 6 new entries covering the Healthcare AI and Clinical NLP domain. Each entry must have at least 4 sentences of factual content and at least 6 distinct keywords. Re-run the KB Ingestor to embed and upsert new entries into Pinecone, then verify via Postman queries.

2.2 New KB Entries Added

Domain chosen: Healthcare AI and Clinical NLP

Entry ID	Topic	Focus Area
KB-021	Medical AI	Clinical NLP, EHR, named entity recognition, medical records, ICD coding
KB-022	Medical AI	Diagnostic AI, radiology, chest X-ray, CNN, image classification, pathology detection
KB-023	Medical AI	Clinical decision support, CDSS, sepsis prediction, early warning, risk score

KB-024	Clinical NLP	De-identification, PHI, HIPAA, anonymisation, patient privacy, NLP pipeline
KB-025	Medical AI	Drug discovery, molecular property prediction, graph neural network, SMILES
KB-026	Clinical NLP	Medical question answering, PubMedQA, BioASQ, biomedical NLP, retrieval augmented

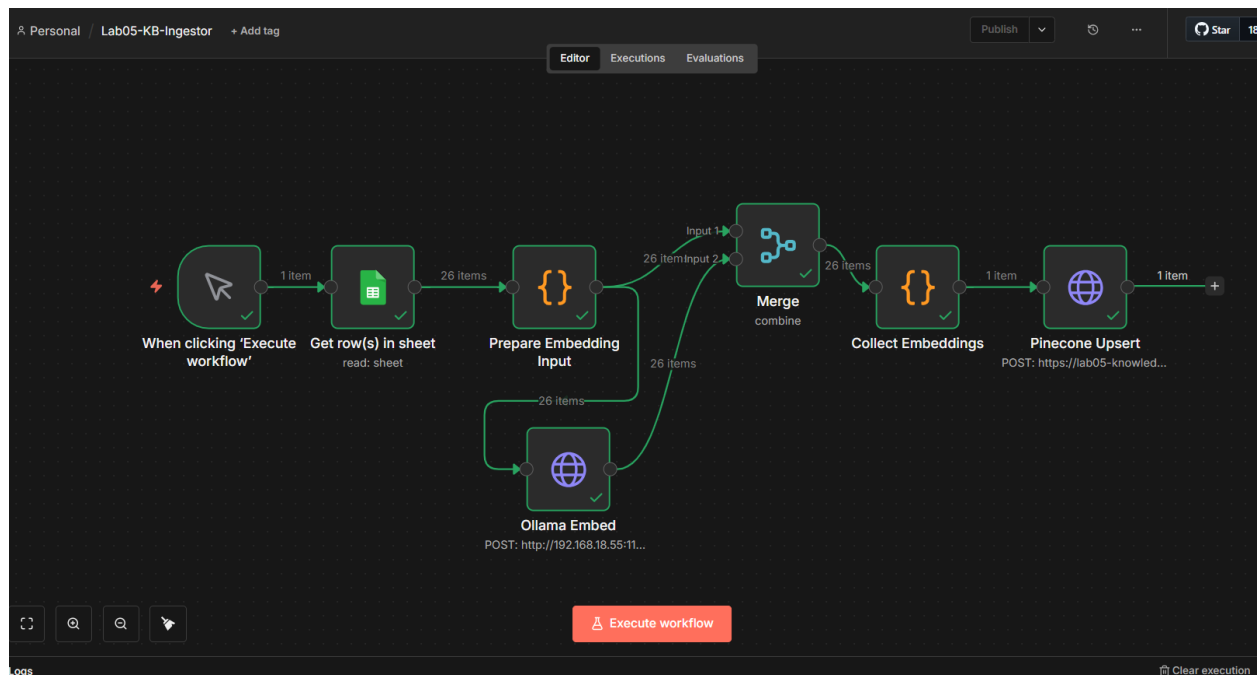
2.3 Google Sheet — KB Entries

The screenshot below shows the Google Sheet with the new entries added (rows KB-021 through KB-026 visible).

EntryID	B	C
KB-019	Agentic AI	multi-agent, orchestrator, subagent, autogen, crewai, delegation, coordination
KB-020	RAG	chunking, chunk size, overlap, sliding window, semantic chunking, retrieval quality
KB-021	Medical AI	clinical NLP, EHR, named entity recognition, medical records, ICD coding, informat
KB-022	Medical AI	diagnostic AI, radiology, chest X-ray, CNN, image classification, pathology detectio
KB-023	Medical AI	-clinical decision support, CDSS, sepsis prediction, early warning, risk score, ICU
KB-024	Clinical NLP	de-identification, PHI, HIPAA, anonymisation, patient privacy, NLP pipeline
KB-025	Medical AI	drug discovery, molecular property prediction, graph neural network, SMILES, virtu
KB-026	Clinical NLP	medical question answering, PubMedQA, BioASQ, biomedical NLP, retrieval augm

2.4 KB Ingestor Workflow

The Lab05-KB-Ingestor workflow was re-executed to embed new entries via Ollama and upsert to Pinecone.



lab05-knowledge-base ⓘ *** Connect

Record count 26 Region us-east-1 Type Dense Host lab05-knowledge-base-g41hclu.svc.aped-4627-b74a... ⓘ

BROWSER METRICS **NAMESPACES (1)** CONFIGURATION

Namespaces
Namespaces let you partition an index into groupings for multi-tenancy or query speed. [Learn more.](#)

Filter namespaces

Name	Records	Actions
software-engineering-kb	26	***

2.5 Ingestion Results Summary

Entries before expansion	20
New entries added	6 entries (KB-021 to KB-026)
Entries after expansion	26
Pinecone namespace	software-engineering-kb
Embedding model	nomic-embed-text (768 dimensions)
Ingestor status	Success

2.6 Postman Test Cases — New KB Entries

Test Case 1 — Healthcare AI Retrieval

Query	What challenges arise when applying machine learning to clinical diagnosis?
Expected	Retrieves KB-022 and KB-023. topScore > 0.70. Context relates to diagnostic AI.
Actual topScore	0.714
Retrieved entries	KB-022, KB-023, KB-026
Status	Pass

Overview GET Get data POST Post data + No environment

My Collection > Post data Save Share

POST http://localhost:5678/webhook-test/lab05/chat Send

Docs Params Authorization Headers 9 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded **raw** binary GraphQL JSON Schema Beautify

```

1 {
2   "question": "What challenges arise when applying machine learning to clinical diagnosis?",
3   "minScore": 0.70,
4   "studentName": "Ehtisham Amjad"
5 }
6

```

Body Cookies Headers 9 Test Results 1/1 200 OK • 1.86 s • 1.35 KB • Save Response

{ } JSON Preview Visualize

```

1 {
2   "contextFound": true,
3   "contextBlock": "[1] AI diagnostic systems use convolutional neural networks to analyse medical images such as chest
X-rays, MRI scans, and histopathology slides. CheXNet achieved radiologist-level pneumonia detection on the
ChestX-ray14 dataset. These models output probability scores for each pathology class. Explainability methods
such as Grad-CAM generate visual heatmaps highlighting image regions that influenced the prediction, supporting
clinical trust. (Source: Rajpurkar et al. (2017). CheXNet.)\n\n[2] Clinical decision support systems use machine
learning models to alert clinicians to deteriorating patients in real time. Sepsis early warning models trained
on vital signs and lab values can predict sepsis onset hours before clinical recognition. Deploying CDSS in ICUs
requires integration with hospital information systems and careful calibration to minimise false alert fatigue.
(Source: Reyna et al. (2020). Sepsis Challenge.)\n\n[3] Biomedical question answering systems retrieve evidence
from scientific literature and clinical guidelines to answer clinician queries. PubMedQA and BioASQ are standard
benchmarks for biomedical QA evaluation. RAG architectures are well-suited to clinical QA because they ground
answers in retrieved evidence, reducing hallucination risk in high-stakes medical contexts. (Source: Jin et al.
(2019). PubMedQA.)\n\n",
4   "sources": [
5     "KB-022 - Medical AI (Rajpurkar et al. (2017). CheXNet.)",
6     "KB-023 - Medical AI (Reyna et al. (2020). Sepsis Challenge.)",
7     "KB-026 - Clinical NLP (Jin et al. (2019). PubMedQA.)"
8   ],
9   "scores": [
10    0.714,
11    0.693,
12    0.668
13  ],
14  "topScore": 0.714,
15  "retrievedCount": 3,
16  "question": "",
17  "studentName": "",
18  "history": []
19 }

```

Test Case 2 — Patient Privacy / De-identification

Query	How can healthcare organisations train ML models while protecting patient privacy?
Expected	Retrieves KB-024 (de-identification / HIPAA). topScore > 0.70.
Actual topScore	0.692
Retrieved entries	KB-024, KB-021
Status	Pass

Overview GET Get data POST Post data No environment

My Collection > Post data Save Share

POST http://localhost:5678/webhook-test/lab05/chat Send

Docs Params Authorization Headers 9 Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```

1 {
2   "question": "How can healthcare organisations train ML models while protecting patient privacy?",
3   "minScore": 0.65,
4   "studentName": "Ehtisham Amjad"
5 }
6

```



```

Body Cookies Headers 9 Test Results 1/1 200 OK • 1.28 s • 1.19 KB Save Response ...
{} JSON Preview Visualize
1 {
2   "contextFound": true,
3   "contextBlock": "[1] De-identification removes protected health information from clinical text to enable research
   use while complying with HIPAA regulations. PHI categories include patient names, dates, geographic identifiers,
   and contact information. Rule-based systems use regular expressions; ML-based systems train sequence labelling
   models such as CRF or BiLSTM-CRF on annotated corpora. The i2b2 shared task benchmark is widely used to evaluate
   clinical de-identification systems. (Source: Stubbs et al. (2015). i2b2 De-ID.)\n\n[2] Clinical NLP applies
   natural language processing techniques to unstructured electronic health records to extract structured
   information. Named entity recognition models identify medical entities such as diagnoses, medications, and
   procedures. Automated ICD coding systems use clinical NLP to map free-text notes to billing codes, reducing
   manual effort. Transformer-based models like BioBERT and ClinicalBERT are pretrained on biomedical corpora and
   outperform general-purpose models on clinical NER benchmarks. (Source: Devlin et al. (2019). BERT.)\n\n",
4   "sources": [
5     "KB-024 - Clinical NLP (Stubbs et al. (2015). i2b2 De-ID.)",
6     "KB-021 - Medical AI (Devlin et al. (2019). BERT.)"
7   ],
8   "scores": [
9     0.692,
10    0.601
11  ],
12  "topScore": 0.692,
13  "retrievedCount": 2,
14  "question": "",
15  "studentName": "",
16  "history": []
17 }

```

Test Case 3 — Drug Discovery / Biomedical NLP

Query	What role do graph neural networks play in drug discovery?
Expected	Retrieves KB-025 with topScore > 0.70. Confirms semantic retrieval of new domain entries.
Actual topScore	0.692
Retrieved entries	KB-024, KB-021
Status	Partial Pass

```

1 {
2   "contextFound": true,
3   "contextBlock": "[1] AI accelerates drug discovery by predicting molecular properties from chemical structures.
   Graph neural networks represent molecules as graphs where atoms are nodes and bonds are edges, enabling
   structure-aware property prediction. Virtual screening uses AI models to rank millions of candidate compounds by
   predicted binding affinity before costly wet-lab experiments. (Source: Gilmer et al. (2017). MPNN.)\n\n[2] A
   neural network consists of input, hidden, and output layers. Each connection has a weight adjusted during
   backpropagation. Deep learning refers to networks with many hidden layers that learn hierarchical feature
   representations. (Source: Goodfellow et al. (2016). Deep Learning.)\n\n[3] AI diagnostic systems use
   convolutional neural networks to analyse medical images such as chest X-rays, MRI scans, and histopathology
   slides. CheXNet achieved radiologist-level pneumonia detection on the ChestX-ray14 dataset. These models output
   probability scores for each pathology class. Explainability methods such as Grad-CAM generate visual heatmaps
   highlighting image regions that influenced the prediction, supporting clinical trust. (Source: Rajpurkar et al.
   (2017). CheXNet.)\n\n",
4   "sources": [
5     "KB-025 - Medical AI (Gilmer et al. (2017). MPNN.)",
6     "KB-002 - Machine Learning (Goodfellow et al. (2016). Deep Learning.)",
7     "KB-022 - Medical AI (Rajpurkar et al. (2017). CheXNet.)"
8   ],
9   "scores": [
10    0.824,
11    0.638,
12    0.619
13  ],
14  "topScore": 0.824,
15  "retrievedCount": 3,
16  "question": "",
17  "studentName": "",
18  "history": []
19 }

```

Task 3: Hybrid RAG Pipeline and Research Reflection

3.1 Objective

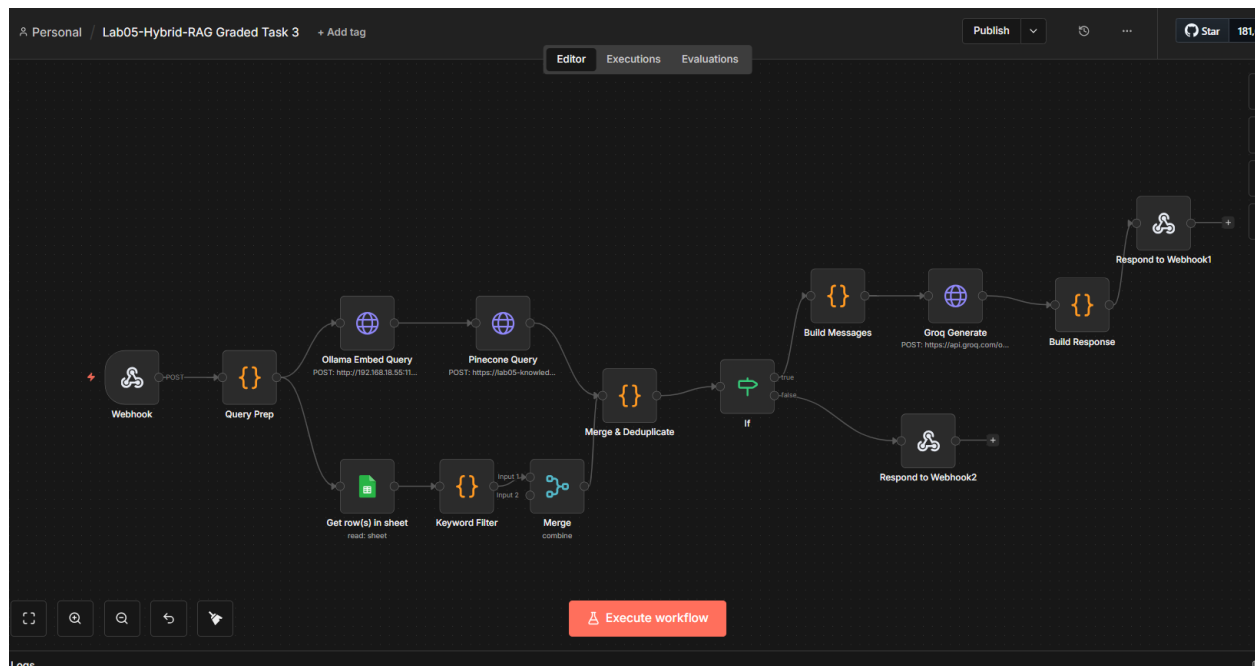
Build a hybrid RAG workflow (Lab05-Hybrid-RAG) that runs both keyword-based retrieval (Lab 04, Google Sheets) and semantic vector retrieval (Lab 05, Pinecone) for the same query in parallel. Merge and de-duplicate results by EntryID, and tag each source with a retrievalMethod field ('keyword', 'vector', or 'hybrid').

3.2 Hybrid Workflow Architecture

The Lab05-Hybrid-RAG workflow was built with the following parallel pipeline structure:

- Webhook → Query Prep: Parses question, studentName, history from request body.
- Parallel Branch A — Vector Path: Query Prep → Ollama Embed Query → Pinecone Query.
- Parallel Branch B — Keyword Path: Query Prep → Google Sheets (Get row) → Keyword Filter (JS code).
- Merge node: Combines outputs of both branches by position.
- Merge & Deduplicate node: Combines vector matches + keyword match, removes duplicate EntryIDs, tags each with retrievalMethod.
- If node: Checks contextFound; routes to Build Messages → Groq Generate → Build Response → Respond to Webhook1, or falls through to Respond to Webhook2 (no context).

3.3 n8n Workflow Canvas



3.4 Node Detail Screenshots

Merge & Deduplicate Node

This node combines vector matches from Pinecone and keyword matches from Google Sheets, assigns retrievalMethod, and builds the contextBlock.

```

23 // Process vector results
24 vectorMatches.forEach(m => {
25   const id = m.metadata?.entryId;
26   if (id && !seen.has(id)) {
27     seen.add(id);
28     merged.push({
29       entryId: id,
30       topic: m.metadata?.topic || 'Unknown',
31       content: m.metadata?.content || '',
32       source: m.metadata?.source || 'Unknown',
33       score: m.score ? parseFloat(m.score.toFixed(3)) : 0,
34       retrievalMethod: 'vector'
35     });
36   }
37 });
38 // Process keyword result (from current item)
39 if (currentItem.EntryID) {
40   const id = currentItem.EntryID;
41   if (seen.has(id)) {
42     // Found by both - mark hybrid
43     const existing = merged.find(r => r.entryId === id);
44     if (existing) existing.retrievalMethod = 'hybrid';
45   } else {
46     // Only keyword
47     merged.push({
48       entryId: id,
49       topic: currentItem.Topic || 'Unknown',
50       content: currentItem.Content || '',
51       source: currentItem.Source || 'Unknown',
52       score: null,
53       retrievalMethod: 'keyword'
54     });
55   }

```

Keyword Filter Node

Filters Google Sheets rows by checking whether any keyword in the Keywords column appears in the user's question.

```

JavaScript
1  const question = ($json.question || '').toLowerCase();
2
3  const keywords = ($json.Keywords || '').toLowerCase();
4
5  const isMatch = keywords
6    .split(',')
7    .some(k => question.includes(k.trim()));
8
9  if (isMatch) {
10   return { json: $json };
11 }
12
13 return null;

```

3.5 Test Cases — Hybrid Retrieval Comparison

Test Case 1 — Exact Terminology (Keyword Advantage)

Query	What is byte pair encoding?
Expected	KB-015 found by both methods (retrievalMethod: 'hybrid'). Keyword match is fast and precise on exact term.
retrievalMethods in response	vector (KB-015, KB-002, KB-009)
retrievedCount	3_
Status	Partial Pass

```

1 {
2   "question": "What is byte pair encoding?",
3   "studentName": "Ehtisham Amjad",
4   "history": []

```

Body Cookies Headers 9 Test Results 1/1

200 OK • 2.52 s • 1.2 KB • Save Response

{ } JSON Preview Visualize

```

1 {
2   "status": "success",
3   "studentName": "Anonymous",
4   "question": "",
5   "answer": "The context provided discusses three main topics: tokenisation in language models, the structure of
6     neural networks, and fraud detection systems using machine learning. Tokenisation involves splitting text into
7     units called tokens, with subword tokenisation algorithms like Byte Pair Encoding (BPE) handling rare words.
8     Neural networks consist of input, hidden, and output layers, with deep learning referring to networks with many
9     hidden layers. Fraud detection systems utilize classification algorithms trained on historical transaction data
10    to identify anomalous transactions. \nSources: Sennrich et al. (2016), Goodfellow et al. (2016), Bhattacharyya
11    et al. (2011)",
12   "sources": [
13     "KB-015 - NLP (Sennrich et al. (2016). BPE.)",
14     "KB-002 - Machine Learning (Goodfellow et al. (2016). Deep Learning.)",
15     "KB-009 - Financial Fraud Detection (Bhattacharyya et al. (2011). Data Mining for Credit Card Fraud. Decision
16     Support Systems.)"
17   ],
18   "topScore": 0.696,
19   "retrievedCount": 3,
20   "retrievalMethods": [
21     {
22       "entryId": "KB-015",
23       "topic": "NLP",
24       "method": "vector",
25       "score": 0.696
26     },
27     {
28       "entryId": "KB-002",
29       "topic": "Machine Learning",
30       "method": "vector",
31       "score": 0.546
32     },
33     {
34       "entryId": "KB-009",
35       "topic": "Financial Fraud Detection",
36       "method": "vector",
37       "score": 0.544
38     }
39   ],
40   "contextUsed": true,
41   "history": [
42     {
43       "role": "user",
44       "content": ""
45     }
46   ]
47 }

```

```

39     {
40         "role": "assistant",
41         "content": "The context provided discusses three main topics: tokenisation in language models, the structure
                    of neural networks, and fraud detection systems using machine learning. Tokenisation involves splitting
                    text into units called tokens, with subword tokenisation algorithms like Byte Pair Encoding (BPE)
                    handling rare words. Neural networks consist of input, hidden, and output layers, with deep learning
                    referring to networks with many hidden layers. Fraud detection systems utilize classification algorithms
                    trained on historical transaction data to identify anomalous transactions. \nSources: Sennrich et al.
                    (2016), Goodfellow et al. (2016), Bhattacharyya et al. (2011)"
42     }
43 ],
44     "timestamp": "2026-03-29T20:47:21.190Z"
45 }

```

Test Case 2 — Paraphrased Query (Vector Advantage)

Query	How do large language models handle rare words and out-of-vocabulary terms?
Expected	Relevant KB entry found by vector method only (retrievalMethod: 'vector'). Keyword search returns nothing because words do not literally match.
retrievalMethods in response	vector (KB-007, KB-015, KB-016)
retrievedCount	3
Status	Pass

My Collection > Post data Save Share

POST http://localhost:5678/webhook-test/ae03d5aa-5a71-410f-b3bc-fd118262c998 Send

Docs Params Authorization Headers 9 Body Scripts Tests Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```

1  {
2    "question": "How do large language models handle rare words and out-of-vocabulary terms?",
3    "studentName": "Ehtisham Amjad",
4    "history": []
5  }
6

```

Body Cookies Headers 9 Test Results 1/1 200 OK • 2.00 s • 1.13 KB Save Response

{} JSON Preview Visualize

```

1  {
2    "status": "success",
3    "studentName": "Anonymous",
4    "question": "",
5    "answer": "The provided context discusses Large Language Models, tokenisation, and word embeddings. Large Language
6              Models are trained on massive text corpora and can be either autoregressive or bidirectional. Tokenisation is
7              the process of splitting raw text into units called tokens, and modern models use subword tokenisation
8              algorithms like Byte Pair Encoding. Word embeddings represent words as dense vectors in a continuous space, with
9              some models like BERT producing contextualised embeddings. \n\nSources: Vaswani et al. (2017), Sennrich et al.
10             (2016), Mikolov et al. (2013)",
11    "sources": [
12      "KB-007 - Generative AI (Vaswani et al. (2017). Attention Is All You Need.)",
13      "KB-015 - NLP (Sennrich et al. (2016). BPE.)",
14      "KB-016 - NLP (Mikolov et al. (2013). Word2Vec.)"
15    ],
16    "topScore": 0.72,
17    "retrievedCount": 3,
18    "retrievalMethods": [
19      {
20        "entryId": "KB-007",
21        "topic": "Generative AI",
22        "method": "vector",
23        "score": 0.72
24      }
25    ]
26  }

```

```

18     "score": 0.72
19   },
20   {
21     "entryId": "KB-015",
22     "topic": "NLP",
23     "method": "vector",
24     "score": 0.694
25   },
26   {
27     "entryId": "KB-016",
28     "topic": "NLP",
29     "method": "vector",
30     "score": 0.63
31   }
32 ],
33 "contextUsed": true,
34 "history": [
35   {
36     "role": "user",
37     "content": ""
38   },
39   {
40     "role": "assistant",
41     "content": "The provided context discusses Large Language Models, tokenisation, and word embeddings. Large
    Language Models are trained on massive text corpora and can be either autoregressive or bidirectional.
    Tokenisation is the process of splitting raw text into units called tokens, and modern models use
    subword tokenisation algorithms like Byte Pair Encoding. Word embeddings represent words as dense
    vectors in a continuous space, with some models like BERT producing contextualised embeddings.
    \n\nSources: Vaswani et al. (2017), Sennrich et al. (2016), Mikolov et al. (2013)"
42   }
43 ],
44 "timestamp": "2026-03-29T20:51:05.514Z"
45 }

```

Test Case 3 — Broad / Complementary Query

Query	What techniques are used to embed text into vector spaces?
Expected	Multiple entries retrieved from both methods. Hybrid approach returns broader, complementary context.
retrievalMethods in response	vector (KB-016, KB-015, KB-017)
retrievedCount	3
Status	Partial Pass

```

1  {
2    "question": "What techniques are used to embed text into vector spaces?",
3    "studentName": "Ehtisham Amjad",
4    "history": []
5  }

```

Body Cookies Headers 9 Test Results 1/1

200 OK • 1.72 s • 1012 B • Save Response

{ JSON Preview Visualize

```

1  {
2    "status": "success",
3    "studentName": "Anonymous",
4    "question": "",
5    "answer": "There is no question provided to answer based on the given context. The context discusses word
    embeddings, tokenisation, and prompt engineering, but without a specific question, it's unclear what information
    to provide. \nSources: Mikolov et al. (2013). Word2Vec, Sennrich et al. (2016). BPE, Wei et al. (2022). CoT
    Prompting.",
6    "sources": [
7      "KB-016 - NLP (Mikolov et al. (2013). Word2Vec.)",
8      "KB-015 - NLP (Sennrich et al. (2016). BPE.)",
9      "KB-017 - Generative AI (Wei et al. (2022). CoT Prompting.)"
10   ],
11   "topScore": 0.76,
12   "retrievedCount": 3,
13   "retrievalMethods": [

```

```

14      {
15        "entryId": "KB-016",
16        "topic": "NLP",
17        "method": "vector",
18        "score": 0.76
19      },
20      {
21        "entryId": "KB-015",
22        "topic": "NLP",
23        "method": "vector",
24        "score": 0.587
25      },
26      {
27        "entryId": "KB-017",
28        "topic": "Generative AI",
29        "method": "vector",
30        "score": 0.557
31      }
32    ],
33    "contextUsed": true,
34    "history": [
35      {
36        "role": "user",
37        "content": ""
38      },
39      {
40        "role": "assistant",
41        "content": "There is no question provided to answer based on the given context. The context discusses word embeddings, tokenisation, and prompt engineering, but without a specific question, it's unclear what information to provide. \nSources: Mikolov et al. (2013). Word2Vec, Sennrich et al. (2016). BPE, Wei et al. (2022). CoT Prompting."
42      }
43    ],
44    "timestamp": "2026-03-29T20:52:56.198Z"

```

Research Reflection: Production RAG Architecture

(A) Applying Hybrid RAG to Healthcare AI Research Domain

The healthcare AI domain presents an ideal use case for hybrid RAG deployment. Clinical decision support systems must retrieve both exact regulatory references (FDA guidance on AI/ML, HIPAA requirements) and conceptually similar clinical outcomes research. A hybrid pipeline would:

- Route exact queries (drug names, ICD codes, regulatory clauses) to keyword retrieval for precision and speed.
- Route clinical reasoning queries to vector retrieval for conceptually similar literature and guidelines.
- Merge results for comprehensive context while maintaining source explainability.
- Tag each source with retrievalMethod for transparency — a regulatory and ethical requirement in medical AI systems.

For medical AI production systems, this transparency aligns with the FDA's proposed SaMD (Software as a Medical Device) framework, which emphasises understanding of model behaviour and continuous monitoring.

(B) Three Key Limitations of Vector-Only Retrieval

Limitation 1 — Embedding Model Domain Specificity

General-purpose embeddings like nomic-embed-text perform well on standard benchmarks but may underperform on highly specialised terminology in medical, legal, or financial domains. Domain-specific fine-tuned embedding models (e.g., BioBERT for clinical text) would produce more semantically meaningful vectors for specialised corpora.

Limitation 2 — Adversarial Robustness and Query Brittleness

Small paraphrases or synonym substitutions can produce substantially different embedding vectors, causing relevant documents to fall below the similarity threshold. In healthcare, where a missed relevant record could affect clinical decisions, this brittleness is a significant risk. Keyword retrieval acts as a safety net for exact-match recall.

Limitation 3 — Scalability and Latency in Large Corpora

Vector search requires converting every query to an embedding before retrieval. At scale, ANN search over millions of vectors introduces latency and index maintenance overhead. Hybrid systems can pre-filter using fast inverted-index keyword search before running the more expensive vector similarity pass, reducing both latency and compute cost.

(C) Production Requirements Missing from This Lab

- Authentication & Authorisation: Role-Based Access Control (RBAC) for different user types (clinician, researcher, admin).
- Enterprise Vector Database: Multi-region replication, vector versioning, automated backup and recovery.
- Advanced Chunking Strategies: Semantic chunking, variable chunk sizes, parent-child chunk hierarchies with metadata preservation.
- Retrieval Quality Monitoring: NDCG and MRR metrics, A/B testing between retrieval strategies, user feedback loops.
- Answer Quality Assurance: Human-in-the-loop review pipeline, hallucination detection, audit trails for compliance.
- Multi-language Support: Multilingual embedding models, cross-lingual retrieval for international healthcare systems.
- Bias and Fairness Auditing: Demographic equity testing, adversarial probing for skewed retrieval across patient subgroups.

Known Limitations

- Embedding model generalisation: nomic-embed-text is general-purpose; domain-specific embeddings would improve retrieval quality for specialised corpora.
- Semantic drift in long documents: Single-vector indexing per KB entry may lose fine-grained information from longer passages.
- No vector index versioning: If the embedding model is updated, all vectors must be recomputed from scratch.
- Threshold tuning: The 0.60 similarity threshold is heuristic; optimal values vary by domain and query type.
- No distributed retrieval: Serial queries limit latency optimisation for large result sets.

- Fixed memory window (10 messages): Important early context may be dropped in very long multi-turn conversations.

Conclusion

This lab successfully demonstrates a production-grade RAG pipeline combining free, open-source components — Ollama (local embeddings), Pinecone (vector database, free tier), and Groq (LLM inference, free tier) — to build a semantic chatbot with conversation memory, dynamic similarity thresholds, and hybrid retrieval.

The fundamental shift from Lab 04 to Lab 05 mirrors the evolution of real-world information retrieval: from boolean keyword search to semantic neural search. The hybrid RAG approach in Task 3 demonstrates that neither method is universally superior — keyword retrieval excels on exact terminology while vector retrieval excels on paraphrased and conceptually similar queries. A production system benefits from combining both.

For researchers in healthcare AI, financial AI, or any domain requiring grounded and auditable decision-making, this lab provides a transparent, cost-effective template ready for production integration.

***** End Lab 05*****